- The Dress Code has to be observed, no warnings will be given, and violators will be asked politely to leave the class and consequently will be marked absent.

Participation

Students are required to attend all classes and read all the assigned material in advance of class (although not necessarily with perfect comprehension). Advanced preparation and class participation are crucial for periods in which we discuss cases. During discussion sessions, the instructor generally keeps track of the insightful and useful comments students make. (Any unproductive contribution is not rewarded)

Reference Material

- Compiler Construction – Principles and Practice By Kenneth C. Louden
- Engineering A Compiler by Keith D.Cooper & Linda Torczon

Calendar of Activities

Weeks	Contents	Activities
1	• Introduction to Compiler Construction • Comparison of interpreters and compilers • Organization • The role of language translation in the programming process • Compiler Construction Basic Concepts • Context of Compiler (Pre-Processor, Assembler • Loader and Link Editor), Phases of compiler • Language translation phases • Front End and Back End	
2	• Lexical Analysis: • The Role of the Lexical Analyzer • Specification of Tokens • Pattern, Lexemes • Recognition of token • Input Buffering, Recognizing More Complex Words • Lexical Error • Specification of Tokens • Regular Expressions (Strings, Numbers, Others)	Assignment-1
3	• Finite Automata • From Regular Expression to Automata • Design of a Lexical-Analyzer Gen. Transition Diagram • Recognition of Reserved Words and Identifier	Quiz-1
4	• Implementing Scanners – hand coded scanners vs automatically generated scanners vs table driven scanners. Ampere's Law and Its Applications	

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	• Implementing Scanners – hand coded scanners vs automatically generated scanners vs table driven scanners. Symbol Table	
5	• Syntax Analysis • Role of the Parser • Context Free Grammars. • Writing a Grammar • Ambiguity Writing Grammar • Eliminating Ambiguity • Elimination left Recursion • Left Factoring • Non Context Free Language Constructs	Quiz-2
6	• Top Down Parsing • Predictive Parsing, Top Down Parsing.	
7	• Recursive Decedent Parsing& Implementation	Assignment-2
8	• Predictive Parsing, LL(1) Parsing& Implementation • Predictive Parsing, LL(1) Parsing& Implementation	
9th Week	**MID TERM**	
10	• Predictive Parsing, LL(1) Parsing& Implementation • Error finding & Recovery	
11	• Error Finding & Recovery • Semantic Analysis • Syntax Directed Definitions • Attributes and Attribute Grammars • Attribute Computations	Quiz -3
12	• Inherited and Synthesized Attributes, Evaluation Orders for SDD's, S-Attributed Definitions, L-Attributed Definitions • Attributes Computation Order • Dependency Graph	Assignment-3
13	• Syntax Directed Translation scheme, Postfix Translation Schemes, SDT's With Actions Inside Productions • Type Checking and Scope	
14	• Type Checking and Scope Implementing L-Attributed SDD's • Intermediate Code Generation • The Three Address Code	Quiz -4
15	• Translation of Expressions • Control Flow, Boolean Expressions, Short-Circuit Code, Control-Flow Translation of Boolean Expression	
16	• Control-Flow Translation of Boolean Expression • Avoiding Redundant Code • Back patching Switch Statement Translation	Assignment-4

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17th Week	**FINAL TERM**	

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